Earth and Planetary Science Letters, 92 (1989) 95–106
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[4]

New paleomagnetic results from Cretaceous sediments near Lisboa (Portugal) and implications for the rotation of Iberia

Armand Galdeano [1], Marie Gabrielle Moreau [1], Jean Pierre Pozzi [2], Pierre Yves Berthou [3] and Jacques André Malod [4]

[1] *Laboratoire de Géomagnétisme et Paléomagnétisme, Institut de Physique du Globe, Université P. et M. Curie, Paris (France)*
[2] *Laboratoire de Géologie, Ecole Normale Supérieure, Paris (France)*
[3] *Laboratoire de Stratigraphie, Université P. et M. Curie, Paris (France)*
[4] *Département de Géologie Dynamique, Université P. et M. Curie, Paris (France)*

Received July 5, 1988; revised version accepted November 9, 1988

Paleomagnetic study of outer platform sediments of Late Jurassic–Early Cretaceous age from Portugal (Lisboa area) leads to characteristic remanent magnetization directions for the Hauterivian–Barremian ($D = 317°$, $I = 58°$) and the Aptian ($D = 344°$, $I = 47°$). The absolute rotation of Iberia between 125 and 110 Ma is consequently of about 27° in an anticlockwise sense. A later rotation of about 14° follows the first one in the same sense from 110 Ma. The total absolute rotation of Iberia from the Hauterivian age determined by paleomagnetism is thus about 41°.

The comparison of declinations measured with paleodeclinations calculated in Lisboa from the Eurasian and African poles of similar age indicates that the main phase of relative rotation of Iberia with respect to Eurasia occurs during the first stage of relative motion (i.e. Hauterivian to Aptian).

The tectonic history of the Bay of Biscay is in good agreement with these two stages of rotation. The total relative rotation with respect to Europe is 34°. This value is consistent with the results previously established by Van der Voo [2,3,25], but we have established that the main phase of rotation (30°) occurs in the Hauterivian–Aptian interval. The movement of Iberia is clearly linked to the movement of Africa in direction and age, but the amount of rotation of Iberia, during the Early Cretaceous, is much larger and implies that Iberia was not a real part of the African plate.

1. Introduction

Since Permian time, the Iberian plate has encountered a counterclockwise rotation, which according to a series of paleomagnetic measurements amounts to 35–40° [1–5]. Measurements on Middle Jurassic rocks give approximately the same value [6–8], indicating that the rotation of Iberia occurred after that time. For the Late Jurassic–Early Cretaceous period, unreliable results have been obtained because of uncertainties in the age determination and of remagnetizations resulting from tectonic deformations [7,8]. Consequently the detailed history of the rotation is not known.

During the Mesozoic, the motion of the small Iberian plate is largely dependent on the motions of the larger ones: Africa, Eurasia and North America. After the beginning of the opening of the North Atlantic it has been dragged by Africa during its eastward motion with respect to Europe and, as a result, has rotated counterclockwise; this rotation has been accompanied by extensional tectonics in the Bay of Biscay. Thus, the tectonic history of the Bay of Biscay should provide information on the Iberia motion.

Three different phases characterize the evolution of the Bay of Biscay region. First, continental rifting was active during the Late Triassic–Liassic, and Late Jurassic–Early Cretaceous. Tectonic activities during the latter period seem to have been the most significant and are characterized by stretching of the continental crust in relation to the formation of numerous tilted blocks [9,10]. West of Lisboa, this phase ended during Tithonian as oceanic crust began to be emplaced between Iberia and Grand Banks (North American plate) [11]. West of Galicia Bank, oceanic breakup

0012-821X/89/$03.50 © 1989 Elsevier Science Publishers B.V.

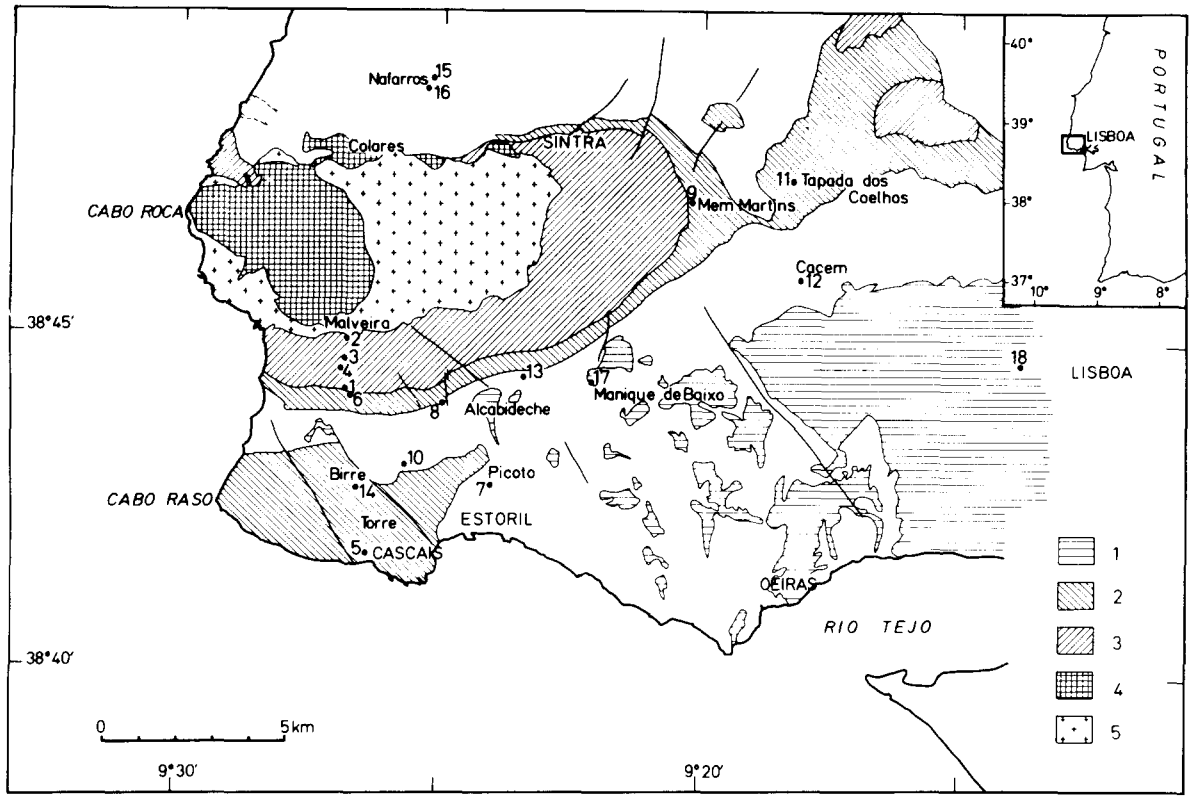


Fig. 1. Simplified geological map of the Lisboa region: Sampling locations have been indicated. Legend: *1* = Lisbon volcanics; *2* = Valanginian to Barremian; *3* = Upper Jurassic; *4* = granite of Sintra; *5* = syenite and gabbro of Sintra.

occurred later during the Late Aptian, as in the Bay of Biscay [9,10]. The second phase corresponds to sea-floor spreading in the Bay of Biscay beginning just after anomaly M0 (Late Aptian–Early Albian) [12,13] and finishing between anomalies 34 and 33 (Campanian) [14,15]. Finally convergence between the Iberian and European plates occurred mainly during the Early Tertiary and resulted into a partial closure of the Bay of Biscay along the North Spanish subduction zone, and into collision in the Pyrenean chain. Olivet [16] has inferred that a counterclockwise rotation of Iberia occurred during this phase from the relative positions of Iberia with respect to Europe given by the oceanic magnetic anomalies. At the end of the Cretaceous, Iberia lay east of its present position with respect to Europe with an azimuthal difference of 5–10°.

The aim of our study is to investigate the problem of the rotation of the Iberian plate during the Late Jurassic–Early Cretaceous, which is the time of beginning of the opening of the Bay of Biscay. The present approach should give a better knowledge not only of the Mesozoic evolution of the Bay of Biscay but also of the regions around the Iberian Peninsula. We carefully selected the region and the sedimentary series to be sampled. A major requirement was to sample well dated rocks in a tectonically stable region of Iberia far away from the Pyrenean and the Betic chains.

The Lisboa region seems to fulfill a lot of these requirements (Fig. 1). It has been slightly deformed during the rifting phases between the Iberian and North American plates, mainly before Early Cretaceous. Some deformation occurred also during the emplacement of the intrusive Sintra Massif, and the Tertiary. Sedimentary facies of outer platform prevailed before the late Barremian. The lithologic and stratigraphic succession is particularly well known [17]. Furthermore, be-

cause of the high density of population, building activity and quarries provided us with numerous fresh outcrops.

2. Geological setting

Mesozoic formations outcrop west of Lisboa along the axis of the Portuguese Mesozoic Rift, active mainly during the Early Liassic and the Late Jurassic (Oxfordian–Kimmeridgian) [18]. The main direction of the rift there is NNE. During the Early Cretaceous, the evolution of this rift stopped completely while the oceanic breakup occurred to the west [11]. In the studied area, Late Jurassic to Early Cretaceous formations are generally tilted southward with higher dips close to the Sintra Massif, which has been emplaced at the end of the Cretaceous [19] and may have induced part of the deformation. Earlier deformations associated with the rifting history are not clearly identified. The Sintra Massif is aligned with the contemporary massifs of Sines Cape and Monchique along a large supposed SSE transcurrent fault [20]. A part of the deformation can also be Tertiary in age. But, this tectonics appears concentrated along old reactivated Hercynian faults, such as in the Serra da Arrabida, 25 km south of Lisboa, or in the Nazare Canyon Thrust, 100 km north of Lisboa. It remains difficult to sort out the deformations due to rifting, to the intrusion of the Sintra Massif and to Tertiary tectonics. However, the general E–W trends of the outcrops as shown on the geological map, and the existence of limited imbricated structures on the northern flank of the Sintra Massif, may indicate that the Tertiary tectonics is responsible for a large part of the southward dips of the Mesozoic series. But in any case, at least south of the Sintra Massif, deformation is low and corresponds to simple tilting of the series. An important feature is the very strong episode of volcanism dated around 70 Ma. Basaltic flows overlies a large zone near Lisbon, around the Sintra Massif and the Tagus river.

We drilled in 18 locations (see Fig. 1). The first locations were located along a N–S cross section perpendicular to the main tectonic structures, where the Late Jurassic and Early Cretaceous outcrops are numerous. The location of the other ones were dependent on the outcroppings. These 18 sites provided 225 cores. The orientation of samples was measured with both magnetic and sun compasses.

3. Paleomagnetic study

Magnetization measurements were carried out on cylindrical standard size samples (2.5 cm in diameter and 2.4 cm length). A fluxgate spinner magnetometer (Molspin, Newcastle, U.K.) was used when the magnetization intensity was higher than 5×10^{-3} A/m, while for lower intensities a cryogenic SQUID magnetometer (CTF, Canada) was used. The contribution of the sample holder has been corrected. Note that the noise level of the cryogenic magnetometer used is experimentally established at about 10^{-11} A m^2 (corresponding to a magnetization of a standard sample of 10^{-6} A/m), i.e. 30 times less than the field produced by the natural remanent magnetization (NRM) of a weakly magnetized sample.

3.1. Rock magnetization

The magnetic properties of the samples were studied to control the stability of the magnetic minerals. These laboratory measurements have been performed on one sample from each site. Examples of some characteristic isothermal remanent magnetization (IRM) acquisition curves are established for a maximum field intensity of 2 T (IRM_{max}). For most of the sedimentary and volcanic samples, magnetic saturation is reached at 0.3 T. This low value indicates the contribution of a low-coercivity phase, probably magnetite or pyrrhotite. However, the samples from sites 10 and 15 do not reach saturation in a field of 2 T, indicating the probable presence of high-coercivity haematite [21,22].

The evolution of the IRM_{max} with reheating has been studied in order to check for changes in magnetic mineralogy (the intensity of IRM_{max} in a field of 2 T is measured after each step of reheating). Results are showed in Fig. 2. Sedimentary samples can be divided into two groups showing typical behaviours. First, the samples from sites 1, 3, 4, and 6, older than Hauterivian, show little evidence of mineralogical transformation below 500 °C (see for examples Fig. 2). At this temperature, the intensity of the IRM_{max} is increased by a ratio which is always lower than 10. Conversely, all samples from sites later than the

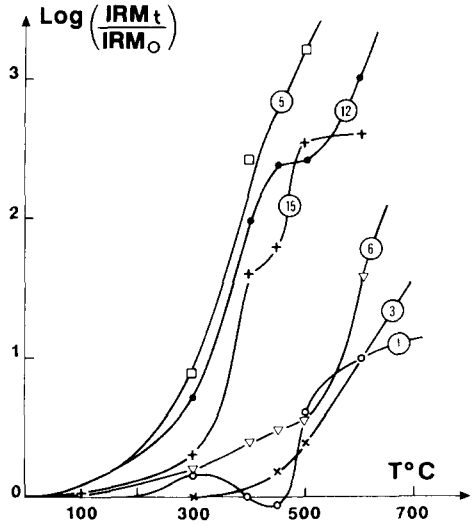


Fig. 2. Evolution of the IRM with respect to reheating temperature. Site numbers are given on each diagram. IRM_t is the IRM measured after reheating and IRM_0 corresponds to IRM_{max} in the text.

Hauterivian age show mineralogical change at a temperature as low as 350 °C. For these samples and those between the temperatures of 350 °C and 500 °C, the intensities of IRM_{max} increase by a ratio exceeding 500 (sites 9, 5, 7, 12, 13, and 15; see also Fig. 2 for examples). This change in the magnetic properties can be linked to the transformation of sulfides or carbonates into oxides. Numerous crystals of pyrite have been observed in polished sections of samples from site 5, along with a few very small grains of magnetite (M. Serment, personal communication).

As a consequence, the range of thermal demagnetization of the post-Hauterivian samples will be less than 400 °C. For the other sites of pre-Hauterivian age, the phase change could probably have already occurred during some geological event and the possibility of a remagnetization must be considered.

3.2. Paleomagnetic results

A number of sample vector demagnetization diagrams (in geographical, i.e. "in-situ" coordinates) are shown in Fig. 3. The results of measurements for each site are listed in Table 1 and displayed (in stereographic projection) in Fig. 4. The following discussion will refer to these figures and to Table 1.

Pre-Hauterivian sites. Site 2 corresponds to a basaltic sill located in the Upper Jurassic sequence. This sill, not reported in the literature, appears to be emplaced conformably into the up-

TABLE 1

Paleomagnetic data

	Site	Tilting	N_t	N_e	α_{95}	K	D_G/I_G	D_S/I_S
Oxfordian	2	55/181	8	8	5.4	108	345/40	243/77
Kimmerigian	3	55/181	9	8	3.3	277	353/30	305/83
	4	60/187	13	13	3.7	138	352/28	265/74
Portlandian	1	37/178	9	8	5.5	101	352/33	343/70
Valanginian	6	65/183	9	8	6.5	76	358/19	322/83
Hauterivian	5	0	11	8	10	35	317/58	317/58
	9a	43/116	10	2	/	/	310/17	323/58
	9b	43/116		7	10.5	33	332/44	45/64
Barremian	8	45/156	7	6	5	132	325/15	315/58
Aptian	10	21/350	11	10	3.5	195	345/75	346/54
	7	4/126	8	6	11	40	338/31	339/33
Albian	13	20/194	7	7	6.8	81	0/36	354/55
	15	7/153	9	8	6.2	81	356/40	358/46
Cenomanian	12	14/159	6	6	6.5	129	357/43	2/56
Upper Cretaceous	17	12/153	13	11	5	85	9/38	15/49

The "tilting" is defined by the dip and the azimuth of geological formations.
N_T = number of processed samples; N_e = number of selected samples.
α_{95} = radius of the circle of 95% confidence; K = precision parameter.
The magnetic directions D_G and I_G are in geographical ("in-situ") coordinates.
The magnetic directions D_S and I_S are in stratigraphic coordinates, after tectonic correction.

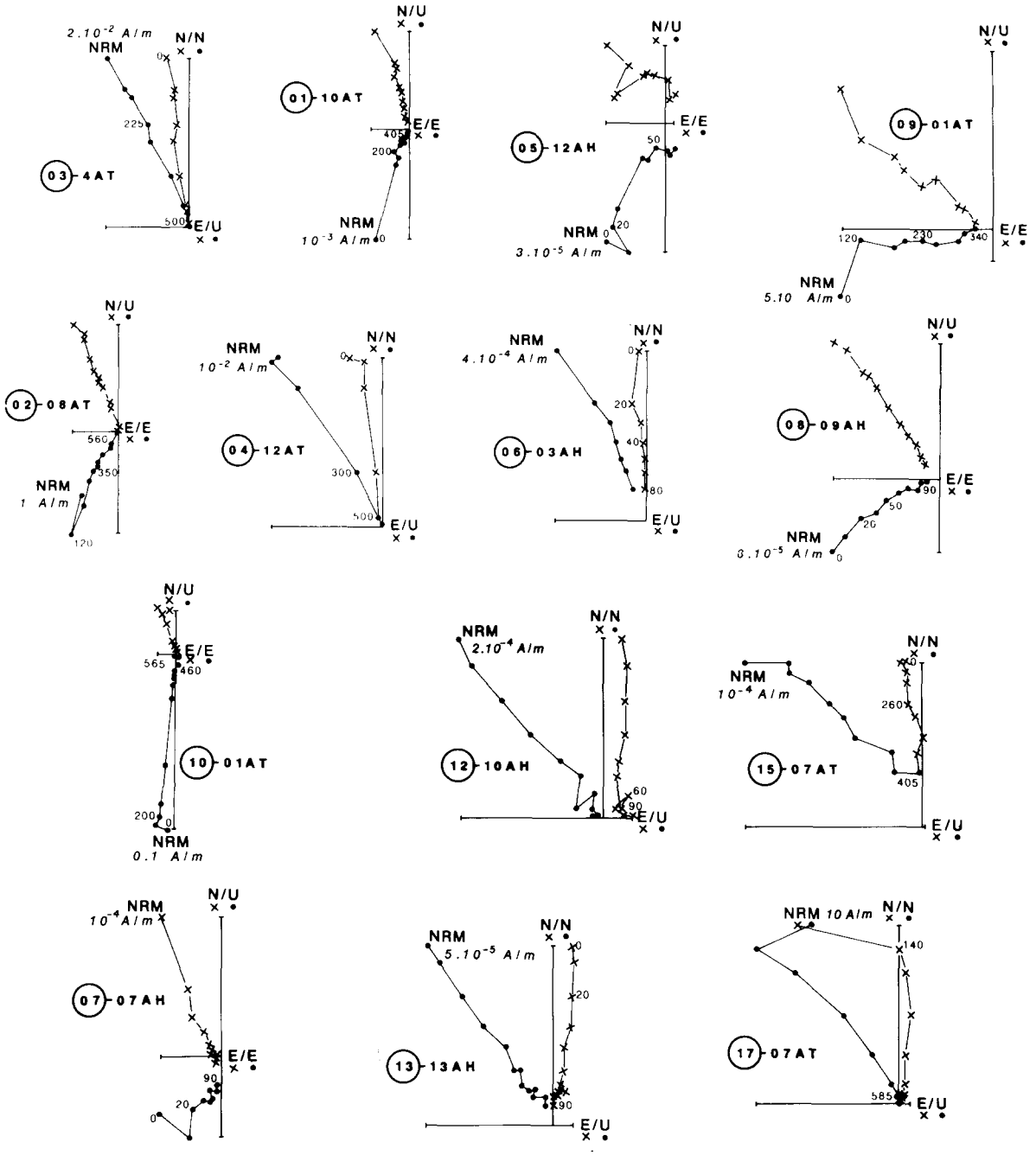


Fig. 3. Orthogonal projection of remanent magnetization vector path during demagnetization in geographic coordinates. The site numbers are given on each diagram (circled numbers). Crosses: projection on the horizontal plane; solid dots: projection on the vertical E–W or N–S plane.

per Oxfordian layers (Carta Geologica de Portugal 1/50000 [23]). The mean intensity of NRM is about 1 A/m and the characteristic direction of magnetization is well defined between 350 and 500 °C.

Sites 3 and 4 are close to each other and

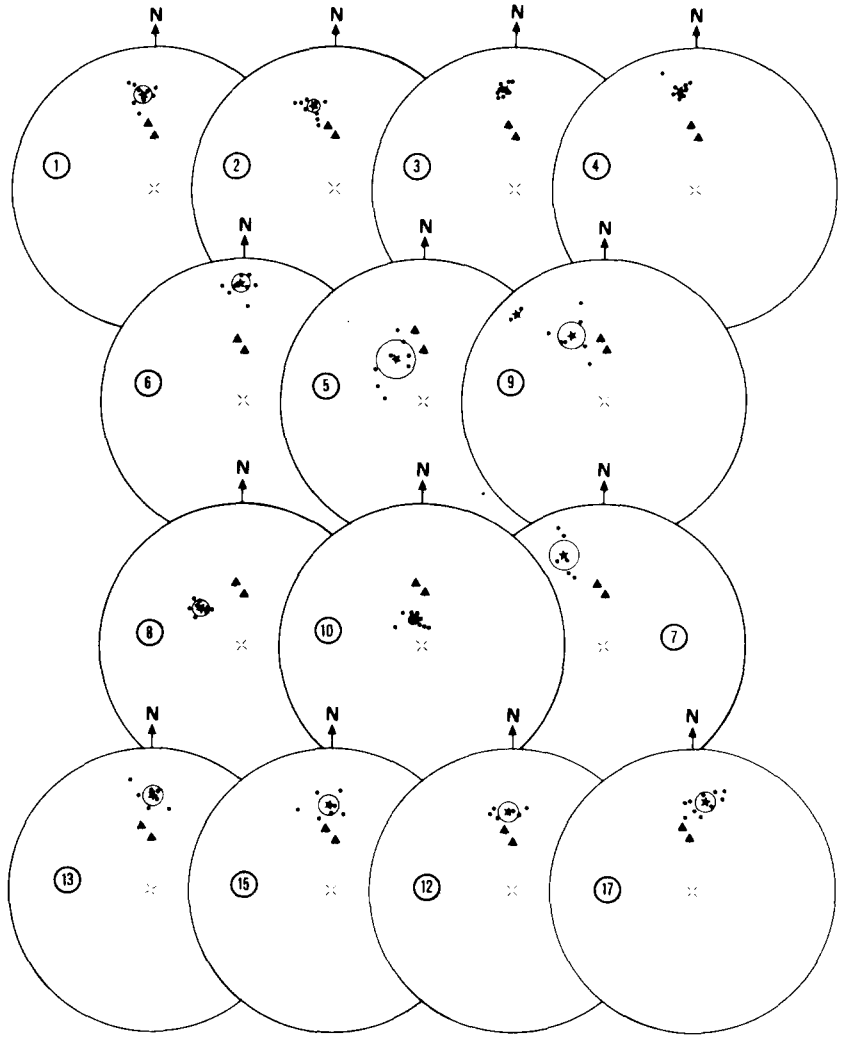


Fig. 4. Stereographic projection (lower hemisphere) of the paleomagnetic directions before tectonic correction (i.e. "in-situ"). Stars: projection of the mean direction for each site. For site 9, directions 9a and 9b have been distinguished. Triangles: projection of the dipole field and of the present magnetic field in Lisboa. The site numbers are given on each projection.

located within limestones of Kimmeridgian age. Demagnetization gives very homogeneous results. The NRM intensities are of the order of 5×10^{-2} A/m and the directions of magnetizations are well clustered between 200° and 500°C.

The samples of sites 1 and 6, respectively of Portlandian and Valanginian age and also located in limestones, exhibit an NRM with an intensity of about 10^{-4} A/m. The characteristic directions are defined between 250° and 400°C. Similar results are obtained by alternating field demagnetization between 30 and 80 mT.

Hauterivian and post-Hauterivian sites. Sites 5, 9, and 14 of Hauterivian age correspond to weakly magnetized samples. For site 14, the directions of magnetization are very scattered. At site 5, despite the very low intensity of the NRM (about 4×10^{-5} A/m), a characteristic direction of magnetization has been determined for most of the samples. Site 9 shows probably an example of local remagnetization: while two samples give the characteristic direction of remagnetization with a low inclination before tectonic correction ("9a"), the other samples define another direction (labeled "9b")

with magnetization vectors which are a little more scattered.

Barremian sites 8 and 11 give NRM intensities of about 10^{-4} A/m. At site 11, magnetization directions before tectonic correction are scattered around the present dipole field, indicating recent remagnetization. In contrast, characteristic directions of site 8 are well defined and grouped for a temperature span from 250° to 450°C or for peak alternating field intensities ranging between 30 and 80 mT.

Sedimentary sites 7 and 10 of Aptian age give NRM intensities which are respectively 2×10^{-4} A/m and 2×10^{-1} A/m. For both sites, the same characteristic magnetization directions are defined between 250° and 450°C or between 10 and 60 mT.

Albian samples from sites 13 and 15, and Cenomanian samples from sites 12 have NRM intensities of about 10^{-4} A/m. Characteristic magnetization directions are defined between 200° and 450°C or between 10 and 60 mT.

Basalts from sites 17 and 18 belong to the Late Cretaceous–Early Tertiary volcanic series of Lisboa. For site 18, the characteristic magnetization has not been recognized (the location of the site, exposed to thunderbolt, must explain this fact). For site 17 the NRM intensity is 10 A/m. Characteristic magnetizations directions are determined between 260° and 500°C.

3.3. *Discussion of the paloemagnetic results*

For the discussion of the paleomagnetic results (Table 1), the sites have been divided into four groups of age: (1) Late Jurassic–Valanginian, (2) Hauterivian–Barremian, (3) Aptian and (4) Albian–Cenomanian.

For the first group, including sites 1, 2, 3, 4, and 6, from Late Jurassic to Valanginian, the fold test [24] is not significant ($k_2/k_1 = 0.66$), indicating probably a post-folding or post-tilting remagnetization.

For the second group (sites 5, 8 and 9), after tectonic correction, the direction "9a" is very close to the directions of the sites 5 and 8. For these three sites the fold test is positive ($k_2/k_1 = 100$), and we can define with a good accuracy ($\alpha_{95} = 4°$) a mean direction for the Hauterivian–Barremian period. The Iberian virtual geomagnetic pole (VGP) for this period is located at 275°E and 58°N.

The Aptian group is represented by sites 10 and 7. The tectonic correction improves clearly the clustering of the directions of magnetization. Computed for all the samples, the fold test is weakly positive ($k_2/k_1 = 2.8$). For the Aptian period, the Iberian VGP is located at 226°E and 84°N.

For the three Albian–Cenomanian sites, 13, 15 and 12, the tectonic directions are relatively close, and the fold test is not significant.

Sites 17 and 18, located on basaltic lava flows of Late Cretaceous–Early Tertiary, were not considered since only the first one gave some representative results.

The remagnetization of the Late Jurassic to Valanginian sites. The samples of pre-Hauterivian age show little or no mineralogical transformation when they are heated in the laboratory (see above). The analysis of these samples provides evidence that mineralogical changes have occurred, for example the transformation of iron sulfides into iron oxides. These transformations may have resulted from a general heating of the region in relation to magmatic intrusions, or, to an activation of the future Atlantic rift (as can be observed with the volcanic sill of site 2). They may have been induced by thermochemical remagnetizations at relatively low temperature (a few hundred degrees).

When drawn on a stereographic diagram, the in-situ directions of magnetization from sites 2, 3, 4 (Late Jurassic), 1, 6 (Valanginian) appears to be coplanar (i.e. located on the same great circle). Furthermore, after tectonic corrections, the directions of magnetization for the Hauterivian–Barremian and Aptian sites are also located in an other plane. But, neither the direction of magnetization related to the Sintra granite [25] nor the present dipole direction are included in these planes. We can assume that the evolution in magnetic direction is caused by a very complex process of acquisition of remagnetization during a probable prerifting uplift of sedimentary beds before the Hauterivian.

Albian Cenomanian magnetizations. One can observe that the fold test is not positive for these series. However, the mean direction for the three

TABLE 2

Summary of results

	Oxfordian/ Valanginian	Hauterivian/ Barremian	Aptian	"Albian/ Cenomanian"	Late Cretaceous
Nb. Sites	5	3	2	3	
D_G; I_G	352; 30	322; 30	340; 59	358; 39	
α_{95}; K_1	8; 91	45; 9	11; 11	6; 383	
D_S; I_S	296; 80	317; 58	344; 47	358; 53	
α_{95}; K_2	10; 59	4; 1000	7; 32	9; 187	
K_2/K_1	0.66	100	2.85	0.5	
VGP:					
long; lat	–	275; 57	226; 77	–	
Paleoposition:					
D ± dD	–	317 ± 7	344 ± 9	→	358 ± 8
Pl ± dPl	–	39 ± 9	28 ± 10	→	22 ± 9

α_{95} = radius of the circle of 95% confidence; K_2/K_1 = McElhinny factor.
The magnetic directions D_G and I_G are in geographical ("in-situ") coordinates.
The magnetic directions D_S and I_S are in stratigraphic coordinates, after tectonic correction.
D and Pl are the paleodeclination and paleolatitude.
The paleoposition in Late Cretaceous time is formed from D_S and I_S estimated for the Albian/Cenomanian and "shifted" in time toward the Late Cretaceous (see text).

sites before tectonic correction (359°, 39.6°) is very close to the direction (358°, 43°) determined by Van der Voo [25] for the Sintra granites (which are 20 Ma younger [19]) or to the magnetic direction (352°, 40°) corresponding to basalts of Lisbon [3] (which are 30 Ma younger) and thus two possibilities can be proposed.

(1) The Sintra granites have been tilted with the surrounding sedimentary series and the direction that we determine is of primary origin. In this case, the magnetization direction for the granite must be corrected in the same way as the sedimentary sites. The mean tectonic directions being N–S, this results in increasing the inclination of about 10°, while the in-situ magnetization direction of the granite is in good agreement with the position of Europe at this time.

(2) A later event at the time of volcanism of Lisbon (about 70 Ma) has remagnetized these sites and the magnetization direction is significant in situ for the end of the Cretaceous or the later Tertiary. One can suppose that the emplacement of widespread basaltic flows can modify the magnetization of shallow sedimentary layers without remagnetizing the lower ones (Hauterivian and Aptian).

In conclusion, although we cannot completely exclude the tilting of the Sintra granite, we prefer the hypothesis of a remagnetization of Albian–Cenomanian sediments at about 70 Ma. However, in both cases the paleodeclinations are very similar.

3.4. Conclusions

The Late Jurassic to Valanginian sites have been remagnetized. Albian–Cenomanian sites have been also probably remagnetized at the epoch of the volcanism of Lisbon. But we can establish good primary magnetization directions for the Hauterivian–Barremian and Aptian sedimentary formations and the corresponding virtual geomagnetic poles can be determined. A recapitulative presentation of these results is shown in Table 2.

4. Geodynamic interpretation and conclusion

The directions of magnetization for the Iberian peninsula between the Hauterivian and the beginning of the Late Cretaceous can be compared to the similar virtual directions computed for European and African paleomagnetic poles at a location corresponding to the present position of Lisboa (lat. 38.75° N, long. 9.4° W). These directions were derived from a careful analysis carried out by Besse [26] from the previous established apparent polar wandering pole (APWP) [27,28].

Our results are compared with the computed paleolatitude and paleodeclination curves (Fig. 5).

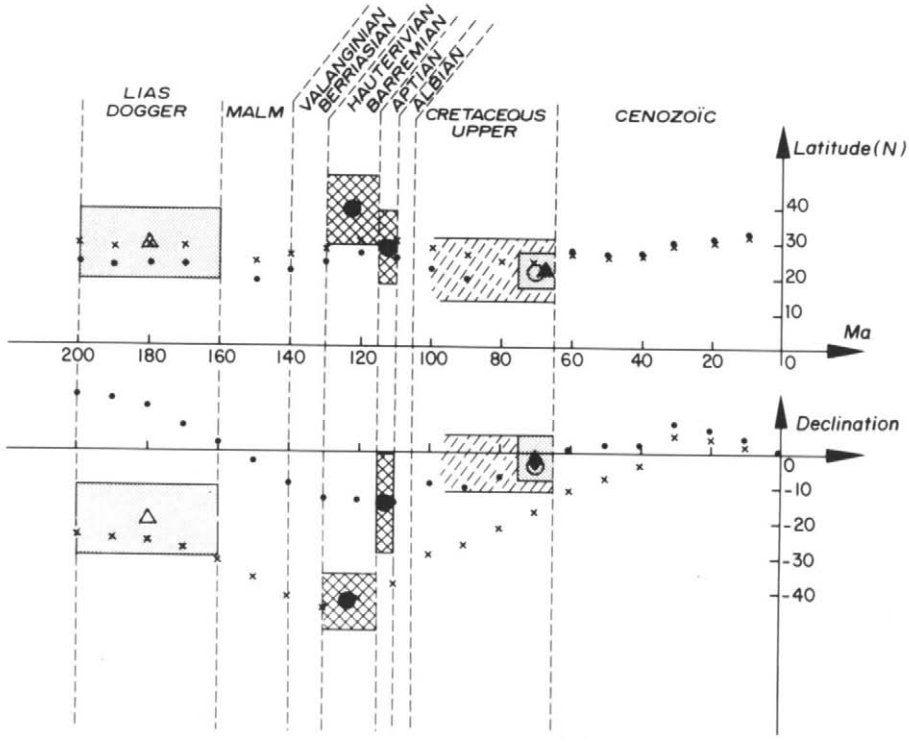


Fig. 5. Paleolatitude and paleodeclination curves. The small crosses (African poles) and small circles (Eurasian poles) indicate the curves of paleolatitude and paleodeclination in a point located in the present position of Lisbon (from [26]). The large circles are the determinations calculated in this paper (close circles: good estimations; open circles; remagnetized sites; see text). Additional determinations from Schott (open triangles) and from Van der Voo (closed triangles) are also shown.

Of course, the post-Aptian points have not been corrected for bedding and the age of their magnetization has been associated with the emplacement of the basalts of Lisbon (about 70 Ma). The point determined by Van der Voo and Zijderveld [3] in these basaltic series and the value given by Schott [6] for Middle Jurassic are also reported.

The diagram corresponding to the paleodeclination can be interpreted in terms of rotations. One can distinguish three main phases:

(1) Rotation before the Hauterivian age. It can be deduced from these diagrams that the movement of Iberia is clearly linked to Africa but that the relative motion with respect to Europe is of weak amplitude.

(2) Rotation within Hauterivian–Aptian interval (125–110 Ma). According to Fig. 5, Iberia rotates in the same sense as Africa but faster. The variation in declination for Iberia is $27° \pm 12°$ during a period of 15 Ma. During the same period the rotation of Eurasia is about $-3°$ (from [26]). This phase corresponds to the main rifting episode and to the formation of the future continental margins in the Bay of Biscay [29]. In this interval, the differential rotation of Iberia with respect to Europe is thus about 30°.

(3) Rotation from Aptian age? The rate of absolute rotation of Iberia decreases to an amplitude of $14 \pm 12°$ for a period greater than 30 Ma. This phase corresponds to the oceanic spreading within the Bay of Biscay which, according to the oceanic magnetic anomalies, has ceased after anomaly 34 (83 Ma). After Cenomanian, our results show that the rotation of Iberia was either terminated or very slow. In the same epoch, the mean rotation rate of Europe is also about 10° and the relative rotation between Iberia and Europe should be estimated at about 4° (from [26]). Obviously, these estimations are poorly resolved with regard to the accuracy of our measurements, but, at the same time, the sea-floor magnetic anomalies of the Bay of Biscay impose a

distensional phase (see below) [29]. Furthermore the relative motion along a direction parallel to the Equator cannot be observed by paleomagnetism. Thus, the observed motions within the Bay of Biscay (from the oceanic magnetic anomalies, for example [30]) can be larger than those determined here.

The total differential rotation between Iberia and Europe is about 34°. This value is consistent with the results previously established by Van der Voo [2,3,25], but we show that the main phase of rotation occurred during the Hauterivian–Aptian interval.

For Early Cretaceous, the paleolatitude calculated from our measurements appears to be too high by about 10° with respect to the ones computed from the Eurasian poles [26]; whereas the N–S motion deduced from kinematic reconstructions [31,32] gives a variation in latitude of about 5°. However, during the Cretaceous period the available data concerning Eurasian and African kinematics are poor [15,32,33]. The APWP curves resulting from combination of poles provided by direct measurements and poles computed from American poles by means of North Atlantic closure models [26–28] are probably the most reliable, but they do not take into account a possible deformation of the Eurasian plate (like that mentioned for a later age by Le Pichon et al. [34]). Note that a similar discrepancy is also depicted by other authors [31,32,35,36] and seems to be characteristic of the Tethysian area.

Our results and those of Schott [6] support the idea that the rotations of Africa and Iberia with respect to Eurasia are interdependent. However, as the rotation of Iberia is faster than that of Africa it is unlikely that Iberia belonged to the African plate between 120 and 95 Ma [32], even if its motion during this latter time span was clearly dependent on the African plate. From our data, a total amount of about 41° of anticlockwise rotation can be estimated between the Hauterivian and the Cenomanian. During this time, the variation in declination of Eurasia is small (about 7°, from [26]).

These values must be compared with the estimated rotations of the Iberian peninsula with respect to Eurasia from Permian/Triassic to Present which range from around 25°, as derived from plate tectonic reconstructions [11,15,31,32], to 34° as measured by paleomagnetism. The discrepancy between the paleomagnetic results and those derived from North Atlantic reconstructions suggests that the Iberian peninsula may have, in part, rotated by itself, as this kind of rotation would not necessarily be recognized in the oceanic magnetic anomalies.

The rotational and latitudinal motions of the Iberian plate deduced from the paleomagnetic study have to be compared with their record in the

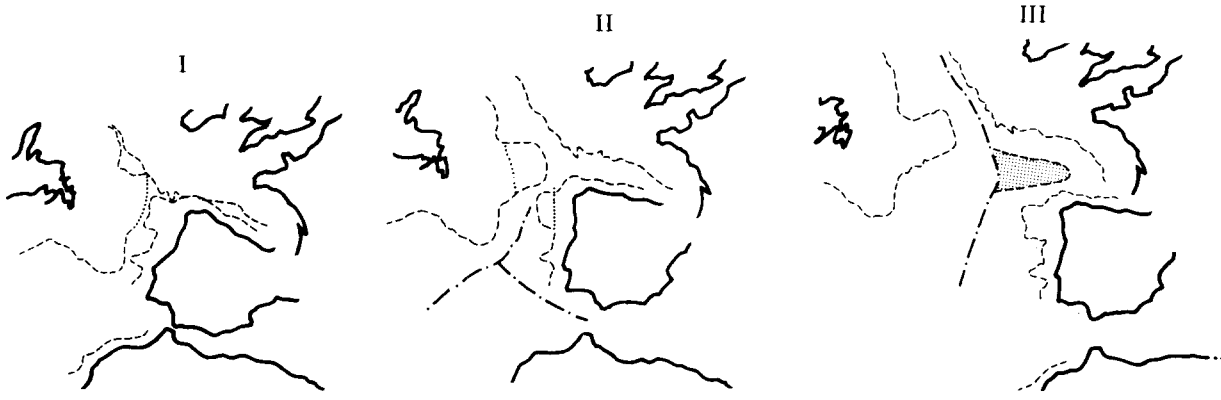


Fig. 6. Schematic representation of motion of Iberia: (I) Late Kimmeridgian; (II) Aptian (about anomaly M0); (III) Campanian (about anomaly 33). Positions I and III are deduced from Atlantic reconstruction. In diagram II, we established the position of Eurasia, North America and Africa in the same way; the position of Iberia in this diagram is deduced from this paper by a rotation of about 27°. Continental rifting appears in the future Bay of Biscay. The rotation of Iberia between stages II and III is weak but oceanic floor is generated in the Bay of Biscay (shaded area in III).

geological features of the Bay of Biscay to specify the timing of the relative movements of Iberia and Eurasia.

The very fast rotation with an amplitude of 27° achieved during Hauterivian–Aptian time can be related to the continental rifting of the same age. We propose that the oceanic spreading of the Bay of Biscay (clearly evidenced by a high-sensibility aeromagnetic survey [37–39]) could originate from a post-Aptian southeastward motion of the Iberian plate accompanied by a weak anticlockwise rotation; this Late Cretaceous movement in latitude is not in contradiction with the evolution of the paleolatitude of Iberia shown Fig. 5 for this period. This spreading is considered to end at anomaly 34 (83 Ma) [14]. This two-fold movement of Iberia is represented schematically Fig. 6.

Acknowledgments

This Research was supported by ATP-CNRS-GGO. Contribution IPGP NS 1036, E.N.S. (Department of Geology) No. 26 and G.E.M.C.O. UA718 No. 472.

References

1 G.J. Van der Lingen, Geology of The Spanish Pyrenees North of Canfranc, Huesca province, Estud. Geol. Inst. Madrid 16, 205, 1960.
2 R. Van der Voo, The rotation of Spain: paleomagnetic evidence from the Spanish Meseta, Palaeogeogr., Palaeoclimatol., Palaeoecol. 3, 393, 1967.
3 R. Van der Voo and J.D.A. Zijderveld, Renewed Paleomametic study of the Lisbon volcanics and implications for the rotation of Iberian Peninsula, J. Geophys. Res. 76, 3913, 1971.
4 K.W. Stauffer and D.H. Tarling, Age of the Bay of Biscay: new paleomagnetic evidence, in: Histoire Structurale du Golfe de Gascogne, 2-II, Technip, Paris, 1971.
5 P.G. Van Dongen, The rotation of Spain: Paleomagnetic evidence from the eastern Pyrenees, Palaeogeogr., Palaeoclimatol., Palaeoecol. 3, 417, 1967.
6 J.J. Schott, R. Montigny and R. Thuizat, Paleomagnetism and potassium argon age of the Messejana Dike (Portugal and Spain): angular limitation to the rotation of the Iberian Peninsula since the Middle Jurassic, Earth Planet. Sci. Lett. 53, 457, 1981.
7 J.J. Schott, Paléomagnétisme des séries rouges du Permien, du Trias et du Crétacé inferieur dans les Chaînes Pyrénéo-Cantabriques et Nord-Ouest Ibériques. Implications géodynamiques, Doc. Thesis, Univ. Louis Pasteur, Strasbourg, 1985.
8 J. Vandenberg, New Paleomagnetic data from the Iberian Peninsula, Geol. Mijnbouw 59, 49, 1980.
9 L. Montadert, D.G. Roberts, G.A. Auffret and P. Guennoc, Rifting and subsidence of the Northern continental margin of the Bay of Biscay, in: Initial Reports of the Deep Sea Drilling Project, Leg 48, p. 1025, U.S. Government Printing Office, Washington, D.C., 1979.
10 Groupe Galice, The continental margin of Galicia and Portugal. Acoustical stratigraphy, dredge stratigraphy and structural evolution, in: Initial Reports of the Deep Sea Drilling Project, Leg 47, Part 2, p. 633, U.S. Government Printing Office, Washington, D.C., 1979.
11 K.D. Klitgord and H. Schouten, Plate kinematics of the Central Atlantic, in: B.E. Tucholke and P.R. Vogt (Editors). The Geology of North America: the Western North Atlantic Region, 1, B.E. Tucholke and P.R. Vogt, eds., Geol. Soc. Am., DNAG Ser., 22, in press, 1988.
12 C.A. Williams, Sea-floor spreading in the Bay of Biscay and its relationship to the North Atlantic, Earth Planet. Sci. Lett. 24, 440, 1975.
13 J.C. Sibuet and W.B.F. Ryan, Site 398: evolution of the west Iberian passive continental margin in the framework of the early evolution of the North Atlantic Ocean, in: Initial Reports of the Deep Sea Drilling Project, Leg 47, Part 2, p. 761, U.S. Government Printing Office, Washington, D.C., 1979.
14 Y. Kristoffersen, Sea floor spreading and the early opening of the North Atlantic, Earth Planet. Sci. Lett. 38, 273, 1978.
15 J.L. Olivet, J. Bonnin, P. Beuzart and J.M. Auzende, Cinématique de l'Atlantique nord et central, 108 pp., Rapp. Sci., C.N.E.X.N.O., Paris, 1984.
16 J.L. Olivet, Nouveau modèle et évolution de l'Atlantique de l'Atlantique nord et central, Doc. Thesis, Université de Paris VII, Paris, 1978.
17 P.Y. Berthou, Résumé synthétique de la stratigraphie et de la paléogéographie du Crétacé moyen et supérieur du bassin occidental portugais, Geonovas 7, 99, 1984.
18 F. Guery, Evolution sédimentaire et dynamique du bassin marginal ouest portugais au Jurassique (Province d'Estremadura de Caldas, secteur da Rainha-Montejunto), Doc. Thesis, Université de Lyon, 1984.
19 M. Bonhomme, F. Mandes and F. Vialette, Ages absolus par la méthode du strontium des granites de Sintra et de Castro Daire au Portugal, C.R. Acad. Sci. Paris 252(21), 3305, 1961.
20 D. Mougenot, Une phase de compression au Crétacé terminal à l'ouest du Portugal: Quelques arguments, Bol. Soc. Geol. Portugal 22, 233, 1981.
21 J. Roquet, Sur les rémanences des oxydes de fer et leur intérêt en géomagnétisme, Ann. Géophys. 10, 226, 1954.
22 D.J. Dunlop, Magnetic properties of fine-particle hematite, Ann. Géophys. 27, 269, 1971.
23 M.M. Ramalho, J. Rey, G. Zbyszewski, F. Moitinho de Almeida, C. Costa and G. Kullberg, Carta Geologica de Portugal na escala de 1/50000, Noticia explicativa da Folha 34c. CASCAIS, Serviços Geologicos de Portugal, 1981.
24 M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J.R. Astron. Soc. 8, 338, 1964.
25 R. Van der Voo, Paleomagnetic evidence for the rotation of the Iberia peninsula, Tectonophysics 7, 5, 1969.

26 J. Besse, Cinématique des plaques et dérive des pôles magnétiques: Evolution de la Téthys, collisions continentales et couplage noyau-manteau, Doc. Thesis, Université de Paris VII, Paris, 1986.
27 M.W. McElhinny, Paleomagnetism and Plate Tectonics, Cambridge University Press, 1973.
28 R. Van der Voo and R.B. French, Apparent polar wandering for the Atlantic-bordering continents: Late Carboniferous to Eocene, Earth Sci. Rev. 10, 99, 1974.
29 J.A. Malod, Comparaison de l'évolution des marges continentales au Nord et au Sud de la Péninsule Ibérique, Doc. Thesis, Univ. Paris VI, Paris, 1982.
30 P. Choukroune, M. Seguret and A. Galdeano, Caractéristiques et évolution structurale des Pyrénées: un modèle de relations entre zone orogénique et mouvement des plaques, Bull. Soc. Géol. Fr. (7) XV, 600, 1973.
31 M. Westphal, M.L. Bazhenov, J.P. Lauer, D.M. Pechersky and J.-C. Sibuet, Paleomagnetic implications on the evolution on the Thethys belt from the Atlantic Ocean to the Pamirs since the Triassic, Tectonophysics 123, 37, 1986.
32 L.A. Savostin, J.C. Sibuet, L.P. Zonenshain, X. Le Pichon and M.J. Roulet, Kinematic evolution of the Tethys Belt from Atlantic Ocean to the Pamirs since the Triassic, Tectonophysics 123, 1, 986.
33 X. Le Pichon, J.-C. Sibuet and J. Francheteau, The fit of the continents around the North Atlantic Ocean, Tectonophysics 38, 169, 1977.
34 X. Le Pichon, F. Bergerat and M.-J. Roulet, Plate kinematics and tectonics leading to the Alpine belt formation; a new analysis, Geol. Soc. Am., Spec. Pap. 218, 111, 1988.
35 A. Gidskehaug, K. Creer and J.G. Mitchell, Paleomagnetism and K-Ar ages of the South-West African basalts and their bearing on the initial rifting of the South Atlantic Ocean, Geophys. J.R. Astron. Soc. 42, 1, 1975.
36 J.C. Briden, A new paleomagnetic result from the Lower Cretaceous of East Central Africa, Geophys. J.R. Astron. Soc. 12, 375, 1967.
37 J.L. Le Mouël and E. Le Borgne, La cartographie magnétique du Golfe de Gascogne, in: Histoire Structurale du Golfe de Gascogne, 2-VI, Technip, Paris, 1971.
38 E. Le Borgne, J.L. Le Mouël and X. Le Pichon, Aeromagnetic survey of South Western Europe, Earth Planet. Sci. Lett. 12, 287, 1971.
39 A. Galdeano, La cartographie aéromagnétique du sud-ouest de l'Europe et de la Région Afar: réalisation, méthodes de traitement, applications géodynamiques, Doc. Thesis, Université de Paris VII, Paris, 1980.